VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai

Department of Computer Engineering)

Department of Computer Engineering

Project Report on

Sannidhi: AI for Proactive and Non-Intrusive Geriatric Health



In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology (B.Tech.) Degree in Computer Engineering at the University of Mumbai

Academic Year 2025-26

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(2025-26)

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

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CERTIFICATE OF APPROVAL

This is to certify that **Neha Mankani, Shraddha Bhagwat, Shambhavi Patil, Supriya Nayak** of Third Year Computer Engineering studying under the University of Mumbai have satisfactorily presented the project on **"Sannidhi: AI for Proactive and Non-Intrusive Geriatric Health"** as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of **Dr. Mrs. Gresha Bhatia** in the year 2025-26.

_____

Date

_____ _____

Internal Examiner **External Examiner**

_____	_____	_____
Project Mentor	**Head of the Department**	**Principal**
Dr. Mrs. Gresha Bhatia	**Dr. Mrs. Nupur Giri**	**Dr. J. M. Nair**

ACKNOWLEDGEMENT

We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

It gives us immense pleasure to express our deep and sincere gratitude to Professor **Dr. Mrs. Gresha Bhatia** (Project Guide) for her kind help and valuable advice during the development of project synopsis and for her guidance and suggestions. We would also like to thank our Alumni Mentor **Mr. Parag Shah** for his invaluable insights

We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair ,** for giving us this valuable opportunity to do this project.

We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

We convey our deep sense of gratitude to all teaching and non-teaching staff for their constant encouragement, support and selfless help throughout the project work. It is a great pleasure to acknowledge the help and suggestion, which we received from the Department of Computer Engineering.

We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

COMPUTER ENGINEERING DEPARTMENT

Course Outcomes for Capstone Project

Learners will be able to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

The rapid expansion of the global geriatric population, coupled with a shrinking caregiver-to-elderly ratio, presents an urgent need for scalable, autonomous eldercare solutions. Traditional monitoring systems suffer from critical limitations: camera-based systems inherently violate patient privacy, while wearable panic buttons suffer from low user compliance and induce "alert fatigue" for caregivers due to high false-positive rates. This project introduces "Sannidhi," a privacy-preserving, device-free eldercare monitoring and autonomous triage system that bridges the gap between passive sensing and active medical intervention.

Sannidhi operates on a novel four-stage methodology: Sense, Think, Act, and Analyze (STAA). The perception layer utilizes distributed ESP32-WROOM nodes to capture ambient WiFi Channel State Information (CSI), acting as an invisible physical tripwire to detect human kinetic anomalies, specifically severe falls. The cognition layer processes these time-series RF distortions via a CNN-GRU-Attention deep learning model hosted locally on an edge device. To eliminate false alarms, the system triggers an Agentic AI core powered by a localized natural language model (Indic-Whisper). This agent autonomously initiates a contextual, multilingual voice check-in (e.g., in Hindi or Marathi) to verify distress before dispatching a critical SOS. Finally, interaction logs are structured through a Medallion Data Warehouse architecture to generate predictive, longitudinal health insights. By combining non-intrusive RF sensing with Agentic AI reasoning, Sannidhi provides a culturally optimized, closed-loop solution that ensures senior safety, preserves user dignity, and significantly reduces the cognitive and emotional burden on caregivers.

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1. Introduction

This chapter introduces the core concepts, motivations, and objectives of the project "Sannidhi" (Advanced Perception & Analytics Flow). It outlines the fundamental drawbacks of existing eldercare monitoring systems and formally defines the problem this project seeks to solve. Furthermore, it highlights the relevance of the proposed system in the context of the growing global geriatric crisis and provides a high-level overview of the Sense-Think-Act-Analyze (STAA) methodology utilized to build this privacy-preserving, autonomous solution.

1.1 Introduction to the project

"Sannidhi" is an advanced, AI-driven eldercare monitoring and triage system designed to provide continuous, privacy-preserving observation without the use of invasive cameras or cumbersome wearables. Utilizing ubiquitous WiFi Channel State Information (CSI), the system operates as an invisible physical tripwire, detecting human kinetic energy and movement patterns, specifically falls. When a high-probability anomaly is detected via edge computing devices, the system employs an Agentic AI core to initiate an autonomous, multilingual voice check-in. This enables real-time verification of distress before escalating alerts to caregivers, thereby creating a closed-loop system that listens, acts, and cares.

1.2 Motivation for the project

The motivation for this project stems from two intersecting global crises: the exponential rise in the geriatric population and the rapidly expanding "caregiver gap." Global data indicates a massive surge in severe falls among the elderly, a trend projected to worsen significantly between 2010 and 2050. Concurrently, sociological shifts in countries like India—specifically the breakdown of the traditional nuclear family structure—have left many seniors living independently without immediate caregiver support. Traditional reactive systems are failing to meet this demand, necessitating a proactive, intelligent, and scalable technological intervention that ensures senior safety while alleviating caregiver anxiety.

1.3 Drawback of the existing system

Current eldercare monitoring solutions are heavily categorized into two paradigms, both of which possess critical flaws:

- **Wearable Devices (Smartwatches, Pendants):** These rely on user compliance. Elderly individuals frequently forget to wear them, forget to charge them, or refuse to wear them due to the stigma associated with physical panic buttons.
- **Camera-Based Systems:** While highly accurate, cameras are fundamentally invasive and violate the privacy and dignity of the user, making them unsuitable for private spaces like bedrooms and bathrooms where falls frequently occur.
- **Passive Infrared (PIR) and Basic Radars:** These offer privacy but lack the granular resolution to differentiate between a simple trip and a critical fall, leading to high false-positive rates. Consequently, caregivers suffer from "alert fatigue," eventually ignoring notifications entirely.

1.4 Problem Definition

To design and develop "Sannidhi," an intelligent, device-free ambient monitoring system that leverages WiFi CSI and Deep Learning (CNN-GRU) to accurately detect falls in real-time. The system must overcome the limitations of false alarms by integrating a localized, Agentic Large Language Model (Indic-Whisper) to autonomously conduct contextual, multilingual voice triage, ultimately generating longitudinal health insights via a Medallion Data Warehouse architecture.

1.5 Relevance of the Project

The project is highly relevant in the modern healthcare and smart-home landscape. It bridges the gap between passive sensing and active intervention. By removing the need for wearables and cameras, it respects patient dignity. Furthermore, by acting as a "first responder" that verbally verifies false alarms, it significantly reduces the cognitive and emotional burden on working caregivers and emergency services, allowing for focused, prioritized care.

1.6 Methodology used

The project utilizes a four-stage, interconnected pipeline termed the **STAA Methodology (Sense, Think, Act, Analyze)**:

- **Sense (Perception Layer):** Distributed ESP32-WROOM nodes capture raw WiFi CSI signal distortions caused by physical movement, acting as an ambient RF sensor.
- **Think (Cognition Layer):** An edge inference model (CNN-GRU-AttNet) on a local host (e.g., Raspberry Pi 4) processes the signal to classify falls, triggering an Agentic AI core.
- **Act (Response Layer):** An autonomous, multilingual voice interface (Indic-Whisper) initiates a verbal check-in (Hindi/Marathi) to verify distress and eliminate false alarms.
- **Analyze (Memory Layer):** Interaction logs are pushed to a Medallion Data Warehouse (Bronze, Silver, Gold layers) to generate predictive insights and detect early cognitive or physical decline.

2. Literature Survey

This chapter reviews the existing literature, industry frameworks, and domain expertise strictly relevant to device-free sensing, fall detection, Medallion data architecture, and Agentic AI in healthcare. It analyzes specific research papers and architectural guidelines to establish current technological benchmarks and summarizes interactions with industry experts to validate the proposed privacy-preserving, edge-computing approach of the Sannidhi project.

2.1 Research Papers

Paper 1. "WiFi CSI based Human Activity Recognition Using Deep Recurrent Neural Network" (ResearchGate)

- **Abstract:** This paper investigates the use of commercial off-the-shelf WiFi devices to capture Channel State Information (CSI) for Human Activity Recognition (HAR). It proposes a Deep Recurrent Neural Network (RNN) architecture to process the time-series CSI data, effectively capturing the temporal dependencies of human movement without requiring wearable sensors.
- **Inference drawn:** This research directly validates the "Sense" and "Think" layers of the Sannidhi methodology. It proves that deep sequential models (like the GRU utilized in our pipeline) can accurately interpret raw RF distortions, providing a strong foundation for our non-intrusive, camera-free fall detection system.

Paper 2. "Redefining Elderly Care With Agentic AI: Challenges and Opportunities" (University of Stirling)

- **Abstract:** This literature explores the paradigm shift from passive AI monitors to "Agentic AI" in geriatric care. It discusses the capabilities of autonomous agents powered by Large Language Models (LLMs) to not only monitor but also reason, interact, and make triaging decisions in real-time, while addressing the ethical and safety challenges of such deployments.
- **Inference drawn:** This paper serves as the theoretical backbone for Sannidhi's "Act" phase. It justifies our departure from simple automated alarms toward an Agentic AI core

that uses contextual reasoning (asking "Are you okay?") to eliminate false positives and provide immediate, localized interaction before alerting human caregivers.

Page 3. "Transforming Healthcare Data Using Medallion Architecture" (CareEvolution)

- **Abstract:** This architectural paper details the implementation of the Medallion data design pattern (Bronze, Silver, Gold layers) specifically within the healthcare sector. It explains how raw, unstructured health logs (Bronze) are progressively sanitized, structured (Silver), and aggregated into analytically ready, clinically actionable insights (Gold).
- **Inference drawn:** This directly structures the "Analyze" phase of our project. It confirms that utilizing a Medallion Data Warehouse is the industry standard for handling longitudinal health data securely, ensuring that Sannidhi's predictive dashboard relies on structured, reliable, "Gold-tier" analytics rather than raw, noisy edge data.

Page 4. "Sahayak: A Senior Citizen Mobile Assistant App" (IRJIET, Vol. 9, Issue 12)

- **Abstract:** This research focuses on the UI/UX and functional requirements of mobile applications designed specifically for senior citizens and their caregivers. It emphasizes high-contrast interfaces, simplified navigation, and essential feature sets required to bridge the digital divide for elderly users.
- **Inference drawn:** While "Sahayak" outlines excellent app design principles, it also highlights the limitations of apps that require active senior participation. We utilized these UI principles for the Sannidhi Caregiver App, but integrated them with our autonomous STAA backend to ensure the senior does not need to actively use a device for their safety to be monitored.

Page 5. Advanced Deep Learning and IoT Frameworks for Healthcare (IEEE / MDPI / OpenReview Synthesis)

- **Abstract:** Recent publications across IEEE and MDPI highlight the optimization of CNNs and attention-based models for edge devices (like Raspberry Pi). Concurrently, OpenReview submissions on audio-processing LLMs emphasize the efficiency of localized speech-to-text models for real-time natural language processing.

- **Inference drawn:** These papers validate our technical stack choices. Running our CNN-GRU-AttNet and localized Indic-Whisper models on the edge is mathematically and computationally feasible, ensuring low latency and strict data privacy by keeping raw audio and CSI data out of the cloud.

2.2 Articles Referred

- **UNFPA India Ageing Report (2023):**



सौरभ गर्ग, भा.प्र.से.
सचिव
Saurabh Garg, IAS
Secretary

भारत सरकार
सामाजिक न्याय और अधिकारिता मंत्रालय
सामाजिक न्याय और अधिकारिता विभाग
Government of India
Ministry of Social Justice & Empowerment
Department of Social Justice & Empowerment

आज़ादी का अमृत महोत्सव

MESSAGE

Department of Social Justice and Empowerment, being the nodal Ministry for the care and welfare of senior citizens, congratulates United Nations Population Fund (UNFPA)-India for bringing out the detailed Report on "Caring for Our Elders: Institutional Responses - India Ageing Report 2023" covering all the major aspects pertaining to institutional care and welfare of senior citizens in the country.

2. As per the Report of the Technical Group on Population Projections-July 2020, the projected population of senior citizens by the year 2036 will be 22.7cr., amounting to 14.9% of the total population of the country, with female senior citizens outnumbering their counterparts. Government of India has been implementing various schemes and programmes for the welfare of senior citizens, especially the indigent ones, for meeting their basic needs of shelter, food, health, clothing and recreation. With advancements in technology, we will be able to serve the senior citizens in a more inclusive way. The Department has been part of preparing Vision-2047 for timely interventions, wherein many existing schemes and programmes will be strengthened and many more initiatives will be undertaken.

3. Government has been always proactive in serving the senior citizens, especially during the COVID-Pandemic period, we have witnessed how every stakeholder, including the Central and State Government, Family, Communities and Civil Society Organisations played their best role during this emergency time. Such an integrated coordination amongst the stakeholders will continue to be needed on a sustained basis for the wellbeing of senior citizens. We also wish good health to senior citizens of the country so as to enable them to remain productive for their families and society at large. We hope that every family builds up a strong inter-generational bonding so as to enable every elderly to respectfully spend the last phase of life with their family.

4. I once again congratulate UNFPA-India for bringing available information regarding Institutional care of senior citizens at one place.

(Saurabh Garg)

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Fig 2.2.1 UNFPA India Ageing Report

This comprehensive report by the United Nations Population Fund highlights the rapid demographic shift in India, projecting that by 2050, the elderly will constitute over 20% of the total population. This data is crucial as it validates the urgent, localized need for scalable eldercare technologies in the face of shrinking traditional family support structures.

- **World Health Organization (WHO) Global Report on Falls:**

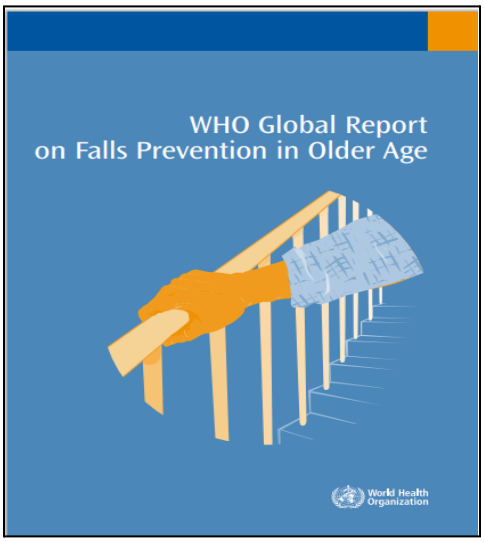


Fig 2.2.2 World Health Organization (WHO) Global Report on Falls

This global demographic assessment establishes falls as a major public health issue and the second leading cause of unintentional injury deaths worldwide. It emphasizes the critical need for immediate post-fall response systems to reduce mortality and severe morbidity.

- **United Nations (UN) World Population Ageing Demographics:**

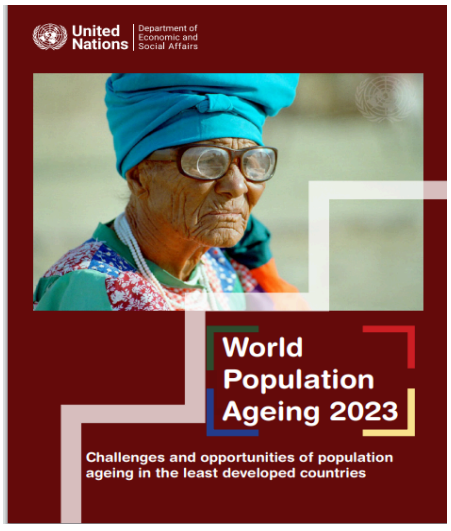


Fig 2.2.3 United Nations (UN) World Population Ageing Demographics

Provides a macro-level perspective on the global shift towards older populations, emphasizing the globally shrinking caregiver-to-elderly ratio, which serves as the primary socio-economic motivation for automating eldercare monitoring.

2.3 Interaction with domain experts

Interaction with Mr. Parag Shah (Apple Inc.): We conducted an architectural and strategic review with Mr. Parag Shah to evaluate the project's execution plan. Rather than focusing solely on the final product, his feedback provided a rigorous, modular engineering framework for the team of four.

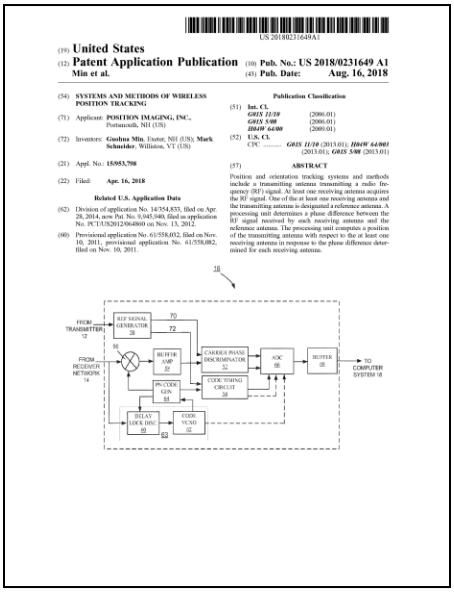
Key Takeaways & Action Items:

- **Identify Unknowns & Validate the Core:** He emphasized that the primary challenge is proving the foundational physics. The immediate action item is to build a proof-of-concept (PoC) prototype focusing strictly on the hardware and CSI pipeline (ESP32 nodes →WiFi capture →CSI conversion) to validate that signal detection works to specification before building the surrounding software.
- **Component Breakdown & Prioritization:** He advised breaking the project into discrete modules (Signal Capture, On-Device Processing, DL Model, Agentic Interaction, Caregiver App). Crucially, he noted that the "Act" and "Analyze" layers are largely "solved engineering problems." Therefore, development priority must be placed on the hardest, unproven pieces: the "Sense" (signal conditioning) and "Think" (inference model) layers.
- **Visualizing Response Efficacy:** To prove the system's value, he recommended creating a temporal timeline that maps the sequence of events (before, during, and after a fall) to explicitly quantify how the STAA methodology drastically shortens response times compared to traditional systems.

2.4 Patent Search

As required by the project index guidelines, the following patents were reviewed to ensure the novelty of the Sannidhi system and to analyze prior art in device-free sensing and automated triage.

1. Title: Device-Free Human Activity Recognition Using WiFi Channel State Information



US 20180231649A1

(19) **United States**

(12) **Patent Application Publication**

Min et al.

(10) **Pub. No.: US 2018/0231649 A1**

(43) **Pub. Date: Aug. 16, 2018**

(54) **SYSTEMS AND METHODS OF WIRELESS POSITION TRACKING**

(71) Applicant: **POSITION IMAGING, INC.**, Portsmouth, NH (US)

(72) Inventors: **Gaobna Min**, Exeter, NH (US); **Mark Schmidtke**, Williston, VT (US)

(21) Appl. No.: **15/953,798**

(22) Filed: **Apr. 16, 2018**

Related U.S. Application Data

(62) Division of application No. 14/354,833, filed on Apr. 28, 2014, now Pat. No. 9,945,940, filed as application No. PCT/US2012/064860 on Nov. 13, 2012.

(60) Provisional application No. 61/558,032, filed on Nov. 10, 2011, provisional application No. 61/558,082, filed on Nov. 10, 2011.

Publication Classification

(51) **Int. Cl.**

G01S 11/10 (2006.01)

G01S 5/00 (2006.01)

H04W 64/00 (2009.01)

(52) **U.S. Cl.**

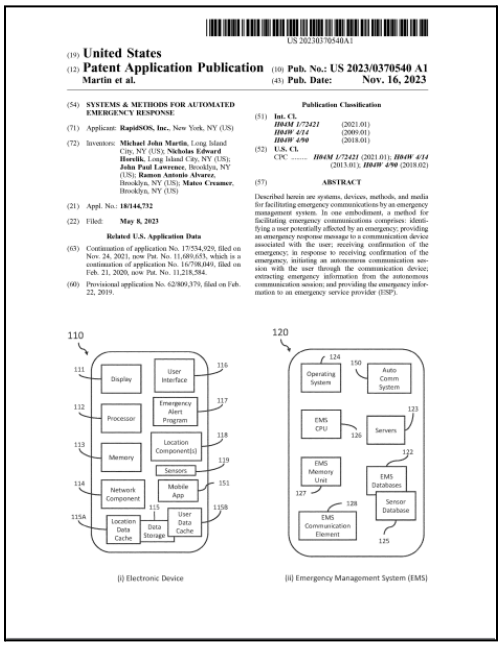
CPC ***G01S 11/10*** (2013.01); ***H04W 64/003*** (2013.01); ***G01S 5/00*** (2013.01)

(57) **ABSTRACT**

Position and orientation tracking systems and methods include a transmitting antenna transmitting a radio frequency (RF) signal. At least one receiving antenna acquires the RF signal. One of the at least one receiving antenna and the transmitting antenna is designated a reference antenna. A processing unit determines a phase difference between the RF signal received by each receiving antenna and the reference antenna. The processing unit computes a position of the transmitting antenna with respect to the at least one receiving antenna in response to the phase difference determined for each receiving antenna.

- **Year:** 2018
- **Summary:** This patent outlines the foundational methodology for utilizing a wireless transmitter and receiver to capture Channel State Information (CSI) data. It describes algorithms for filtering phase shifts and amplitude variations to detect human presence and basic activities in a designated space without the subject wearing any tags or devices.
- **Inference:** This patent confirms that the core physics of our "Sense" layer (using WiFi subcarriers for physical detection) are legally established and viable. Sannidhi builds upon this by adding localized Deep Learning and Agentic AI reasoning, which this patent does not cover.
- **US Patent Link:** https://patents.google.com/patent/US20180231649A1

2. Title: Automated Emergency Triage and Communication System via Natural Language Processing



US 20230370540A1

(19) **United States**

(12) **Patent Application Publication**
Martin et al.

(10) **Pub. No.: US 2023/0370540 A1**
(43) **Pub. Date: Nov. 16, 2023**

(54) **SYSTEMS & METHODS FOR AUTOMATED EMERGENCY RESPONSE**

(71) Applicant: **RapidSOS, Inc.**, New York, NY (US)

(72) Inventors: **Michael John Martin**, Long Island City, NY (US); **Nicholas Edward Horsik**, Long Island City, NY (US); **John Paul Lawrence**, Brooklyn, NY (US); **Ramon Antonio Alvarez**, Brooklyn, NY (US); **Mateo Creamer**, Brooklyn, NY (US)

(21) Appl. No.: **18/144,732**

(22) Filed: **May 8, 2023**

Related U.S. Application Data

(63) Continuation of application No. 17/554,929, filed on Nov. 24, 2021, now Pat. No. 11,689,653, which is a continuation of application No. 16/798,049, filed on Feb. 21, 2020, now Pat. No. 11,218,584.

(60) Provisional application No. 62/809,370, filed on Feb. 22, 2019.

Publication Classification

(51) **Int. Cl.**
H04M 1/72421 (2021.01)
H04W 4/14 (2009.01)
H04W 4/90 (2018.01)

(52) **U.S. Cl.**
CPC ***H04M 1/72421*** (2021.01); ***H04W 4/14*** (2013.01); ***H04W 4/90*** (2018.02)

(57) **ABSTRACT**

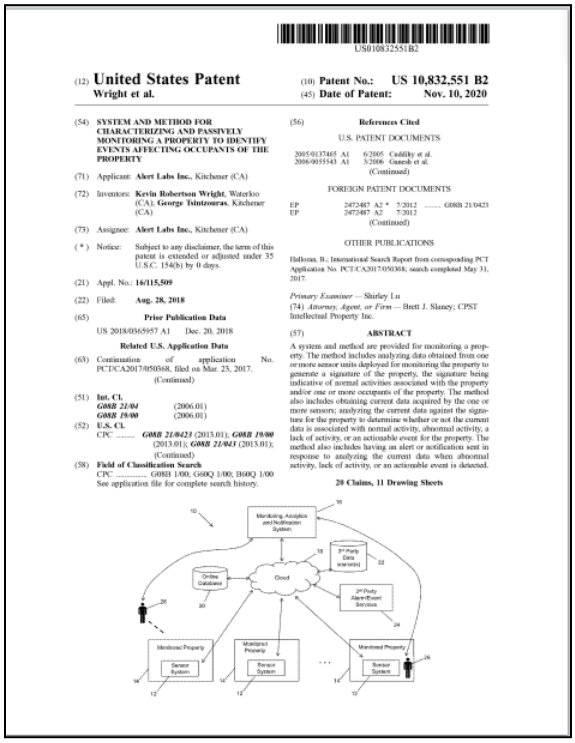
Described herein are systems, devices, methods, and media for facilitating emergency communications by an emergency management system. In one embodiment, a method for facilitating emergency communications comprises: identifying a user potentially affected by an emergency; providing an emergency response message to a communication device associated with the user; receiving confirmation of the emergency; in response to receiving confirmation of the emergency, initiating an autonomous communication session with the user through the communication device; extracting emergency information from the autonomous communication session; and providing the emergency information to an emergency service provider (ESP).

(i) Electronic Device

(ii) Emergency Management System (EMS)

- **Year:** 2021
- **Summary:** A system designed to intercept emergency signals and utilize a conversational AI agent to communicate with the user. The system determines the severity of the emergency through keyword extraction and subsequently routes the data to the appropriate human caregiver or medical service.
- **Inference:** This demonstrates industrial prior art for the conversational "Act" phase of our project. Sannidhi's novelty lies in triggering this NLP triage system strictly via device-free RF sensing and utilizing edge-based, localized Indian language models (Indic-Whisper) rather than cloud-based APIs.
- **European Patent Link:** https://worldwide.espacenet.com/patent/search/family/072671391/publication/EP3812345A1

3. Title: Method and System for Fall Detection Using Wireless Signals and Machine Learning



US010832551B2

(12) **United States Patent**
Wright et al.

(10) **Patent No.: US 10,832,551 B2**
(45) **Date of Patent: Nov. 10, 2020**

(54) **SYSTEM AND METHOD FOR CHARACTERIZING AND PASSIVELY MONITORING A PROPERTY TO IDENTIFY EVENTS AFFECTING OCCUPANTS OF THE PROPERTY**

(71) Applicant: **Alert Labs Inc.**, Kitchener (CA)

(72) Inventors: **Kevin Robertson Wright**, Waterloo (CA); **George Tsintzouras**, Kitchener (CA)

(73) Assignee: **Alert Labs Inc.**, Kitchener (CA)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **16/115,509**

(22) Filed: **Aug. 28, 2018**

(65) **Prior Publication Data**

US 2018/0365957 A1 Dec. 20, 2018

Related U.S. Application Data

(63) Continuation of application No. PCT/CA2017/050368, filed on Mar. 23, 2017. (Continued)

(51) **Int. Cl.**
G08B 21/04 (2006.01)
G08B 19/00 (2006.01)

(52) **U.S. Cl.**
CPC ..... ***G08B 21/0423*** (2013.01); ***G08B 19/00*** (2013.01); ***G08B 21/043*** (2013.01); (Continued)

(58) **Field of Classification Search**
CPC G08B 1/00; G08Q 1/00; H60Q 1/00
See application file for complete search history.

(56) **References Cited**

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2006/0055543 A1 3/2006 Ganesh et al.
(Continued)

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EP 2472487 A2 7/2012
(Continued)

OTHER PUBLICATIONS

Halloran, B.; International Search Report from corresponding PCT Application No. PCT/CA2017/050368; search completed May 31, 2017.

Primary Examiner — Shirley Lu
(74) *Attorney, Agent, or Firm* — Brett J. Slaney; CPST Intellectual Property Inc.

(57) **ABSTRACT**

A system and method are provided for monitoring a property. The method includes analyzing data obtained from one or more sensor units deployed for monitoring the property to generate a signature of the property, the signature being indicative of normal activities associated with the property and/or one or more occupants of the property. The method also includes obtaining current data acquired by the one or more sensors; analyzing the current data against the signature for the property to determine whether or not the current data is associated with normal activity, abnormal activity, a lack of activity, or an actionable event for the property. The method also includes having an alert or notification sent in response to analyzing the current data when abnormal activity, lack of activity, or an actionable event is detected.

20 Claims, 11 Drawing Sheets

- **Year:** 2020
- **Summary:** This patent details a system that extracts spatial and temporal features from wireless signal reflections to identify fall events. It explicitly mentions the use of basic machine learning classifiers to distinguish falls from normal daily activities like walking or sitting.
- **Inference:** While this patent covers basic machine learning for RF fall detection, it does not account for the autonomous elimination of false positives. Our integration of a "Think-to-Act" pipeline—where the system verbally verifies the fall before sending an alert—bypasses the limitations described in this prior art.
- **US Patent Link:** https://patents.google.com/patent/US10832551B2

3. Requirements of Proposed System

3.1 Functional Requirements

The Sannidhi framework must seamlessly execute a series of interconnected operations to ensure the safety of elderly users:

- **Continuous Ambient Sensing:** The system is required to continuously and passively monitor the environment by capturing ambient WiFi Channel State Information (CSI) signals, detecting spatial movements without requiring user interaction or wearable devices.
- **Local Fall Detection:** Upon capturing these signals, the local edge node must process the multi-dimensional tensors using a deployed deep learning model to accurately identify severe falls in real-time.
- **Proactive Verification:** If an anomaly is detected, the system must autonomously initiate a proactive, two-way voice check-in using regional languages (Hindi/Marathi) to verify the user's condition.
- **Emergency Alerting:** Based on verbal verification or a lack thereof, the system must trigger a critical emergency protocol, dispatching SOS notifications and precise location data to a designated caregiver's mobile application.
- **Longitudinal Health Tracking:** The framework must continuously route all interaction logs and daily mobility metrics to a secure data warehouse to generate predictive weekly reports on the user's health trends.

3.2 Non-Functional Requirements

To ensure user trust and clinical utility, the system must adhere to strict non-functional standards:

- **Absolute Privacy:** As the cornerstone of this architecture, the system must remain entirely camera-free and process all sensitive sensor and audio data locally on the edge device, ensuring no raw surveillance data is ever streamed to the cloud.

- **High Accuracy:** The activity recognition model must maintain a high operational standard, targeting an accuracy rate of over 90% even in Non-Line-of-Sight (NLoS) environments where walls or furniture obscure the subject.
- **Low Latency:** Edge inference must process sliding windows of data with minimal delay to provide real-time responses and faster speeds during critical emergencies.
- **Decoupled Architecture:** The overarching software architecture must be highly modular, ensuring that a failure in the alerting backend does not compromise the continuous local sensing and inference capabilities.

3.3 Constraints

Operational deployment is bound by specific hardware and environmental constraints:

- **Hardware Limitations:** Because all primary deep learning inference and local natural language processing occur on a single Raspberry Pi 4, the computational complexity and thermal output of the models must be strictly managed to prevent throttling.
- **Signal Dependency:** The core sensing modality relies entirely on stable ambient WiFi signals; severe electromagnetic interference or a complete router outage will temporarily degrade physical monitoring capabilities.
- **Acoustic Environment:** The effectiveness of the autonomous voice verification requires the ambient noise levels in the room to be low enough for the local microphone to accurately capture the senior's verbal responses.

3.4 Hardware & Software Requirements

The physical and digital infrastructure is built upon lightweight, ubiquitous components:

- **Hardware:**
    - ESP32-WROOM modules configured as transmitters and receivers to capture CSI data.
    - Raspberry Pi 4 acting as the centralized local edge host.
    - Standard USB microphones and speakers for agentic voice interactions.

- **Software & Frameworks:**
    - TensorFlow and Keras ecosystems for model development and edge deployment.
    - LangChain utilizing Retrieval-Augmented Generation principles for autonomous reasoning.
    - IndicWhisper to handle Hindi and Marathi speech recognition.
    - PostgreSQL database for the analytical backend.
    - React Native or Flutter for the cross-platform caregiver interface.

3.5 Techniques utilized till date for the proposed system

Several advanced methodologies have been integrated to build the foundational proof-of-concept:

- **Device-Free Activity Recognition:** Translating physical human movements into measurable CSI subcarrier variations by interpreting the multipath propagation of radio frequency signals.
- **Domain Adaptation via Generative Models:** Deploying a conditional Wasserstein GAN with Gradient Penalty (cWGAN-GP) to generate synthetic CSI samples, which effectively simulates various spatial environments during training and prevents overfitting.
- **Agentic NLP Orchestration:** Transitioning from a traditional, binary alarm system into an empathetic framework capable of contextual reasoning and two-way verbal verification.
- **Medallion Data Architecture:** Implementing a tiered strategy (Bronze, Silver, Gold) to ensure high-frequency, chaotic telemetry is systematically refined into structured, longitudinal health insights.

3.6 Tools utilized till date for the proposed system

- **Python:** The primary language handling everything from initial signal sanitization to the orchestration of the generative pipelines.

- **TensorFlow Lite:** Crucial for quantizing and optimizing the complex neural networks so they run efficiently within the constrained edge environment.
- **MQTT Protocol:** Functions as the central message broker for asynchronous communication, allowing sensing nodes to reliably trigger the NLP agent.
- **PostgreSQL:** Provides the relational data management backbone, executing queries required to transition raw event logs into actionable predictive analytics.

3.7 SDGs mapped with the proposed project and justification

- **Goal 3: Good Health and Well-being:** By directly mitigating the severe morbidity associated with the "Long Lie" phenomenon in isolated elderly populations and tracking health declines, it fundamentally extends the period of safe, independent living.
- **Goal 9: Industry, Innovation, and Infrastructure:** The project drastically reduces the need for intrusive hardware installations by repurposing standard home WiFi networks into healthcare monitoring radars, pioneering privacy-preserving technologies in the edge computing sector.

4. Proposed Design

4.1 Block diagram representation of the proposed system with explanation

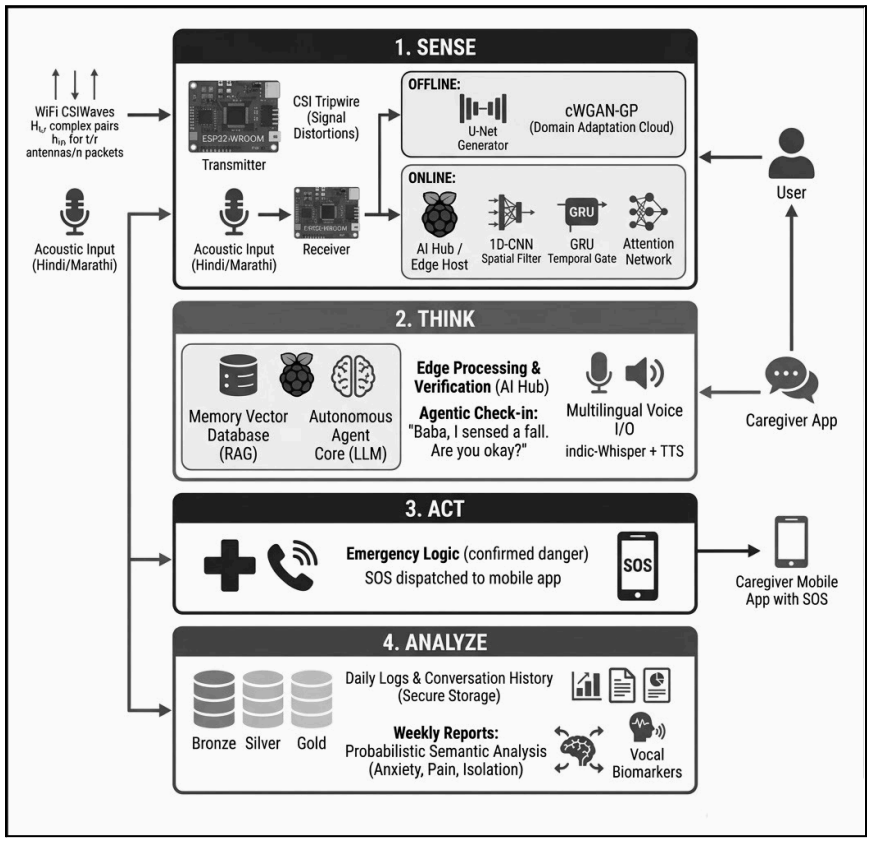


Fig. 4.1.1 Block diagram

The systemic logic of Sannidhi operates on a continuous Sense-Think-Act-Analyze (STAA) methodology:

- **Sense:** Localized ESP32 nodes capture distortions in ambient WiFi CSI signals, functioning as an invisible and privacy-preserving physical tripwire.
- **Think:** Raw data is transmitted to the local Raspberry Pi, where a customized neural network evaluates signal variations. If a fall pattern is detected, it triggers the Agentic AI core to initiate verbal verification.

- **Act:** Dependent on the context of the user's response (or lack thereof), the system bypasses local logic to dispatch an instant SOS notification and geolocation metadata via cloud messaging to the caregiver's application.
- **Analyze:** Daily routines and verified incidents are securely routed to the PostgreSQL data warehouse, where batch jobs generate probabilistic reports mapping out long-term behavioral trends.

4.2 Modular diagram representation of the proposed system with explanation

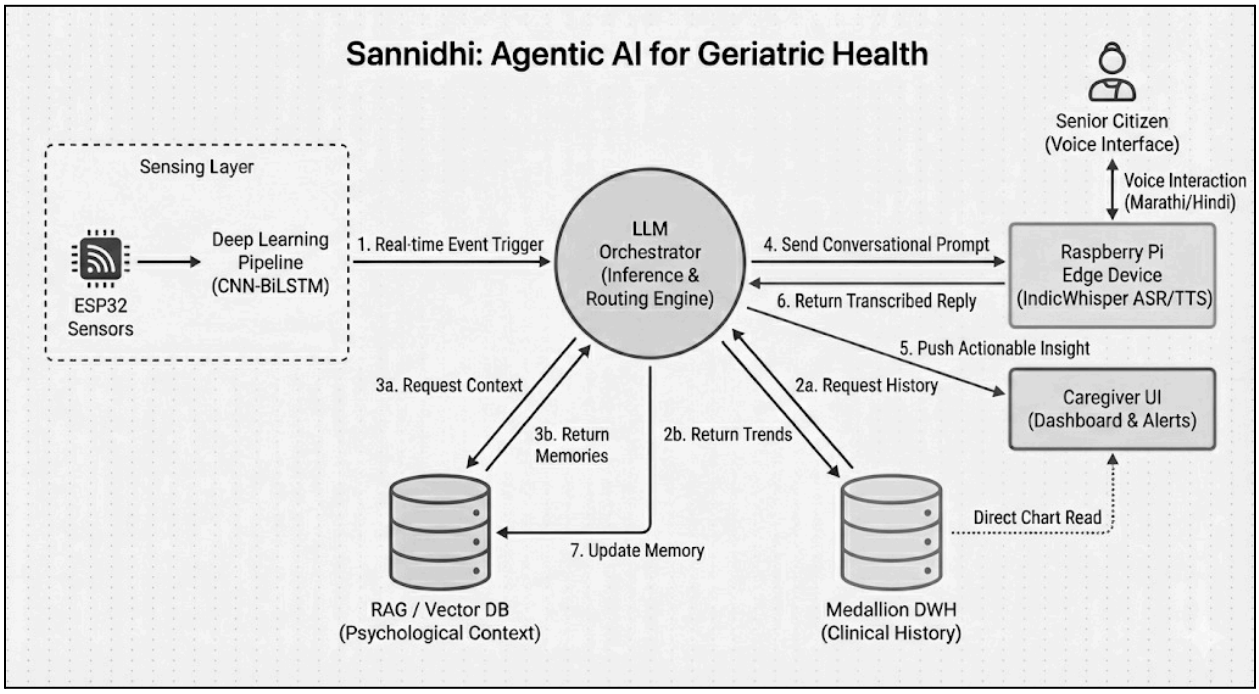


Fig. 4.2.1 Modular diagram

The framework is designed as a composition of four highly independent modules:

- **Signal Acquisition Module:** Operates at the hardware boundary, managing raw CSI extraction, applying phase sanitization, suppressing static noise, and formatting data into overlapping sliding windows.

- **Edge Inference Module:** Encapsulates the optimized deep learning model responsible for calculating activity probabilities and generating structured JSON payloads when anomalies occur.
- **Agentic Orchestration Module:** Subscribes to MQTT alerts, dynamically constructs conversational prompts, interfaces with speech-to-text engines, and handles the logical routing of the emergency response.
- **Data Analytics Module:** Comprises the tiered data warehouse, managing the ingestion of raw logs, the cleaning of structured data, and the execution of the predictive insights layer.

4.3 Design of the proposed system – Data Flow / Activity Diagram

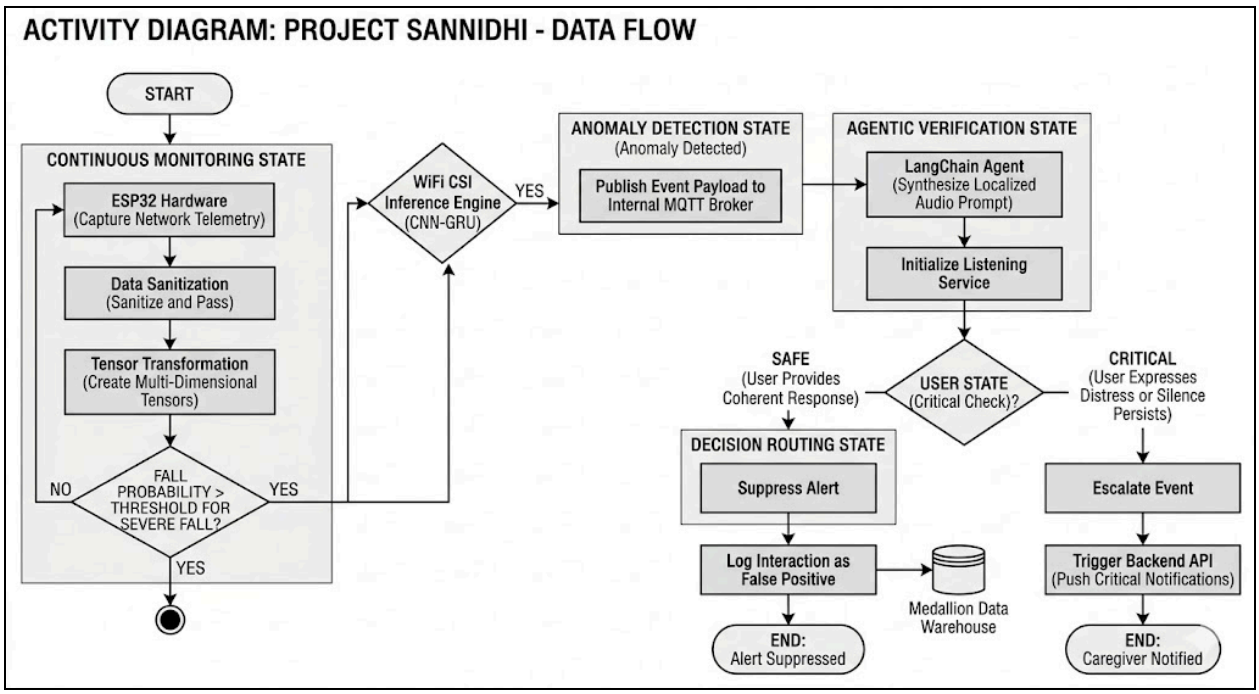


Fig. 4.3.1 Block diagram

The system's data flow transitions through four distinct operational states:

1. **Continuous Monitoring State:** ESP32 hardware consistently captures network telemetry, which is sanitized and passed as multi-dimensional tensors into the inference engine.
2. **Anomaly Detection State:** If the model's output probabilities breach a predefined threshold for a severe fall, an event payload is published to the internal MQTT broker.
3. **Agentic Verification State:** The LangChain-powered agent synthesizes a localized audio prompt inquiring about the user's safety and initializes the listening service.
4. **Decision Routing State:**
    - *Safe:* If the user provides a coherent response, the flow suppresses the alert and logs the interaction as a false positive.
    - *Critical:* If distress is expressed or acoustic silence persists, the flow escalates the event, triggering the backend API to push critical notifications.

4.4 Algorithms utilized in the existing systems

The architecture relies on a specialized suite of algorithms to bridge the gap between raw physics and contextual understanding:

- **cWGAN-GP (Conditional Generative Models):** Deployed strictly during the offline training phase to mathematically simulate variations in room geometry, creating a robust, augmented dataset that prevents environmental overfitting.
- **CNN-GRU-Attention Network:** The primary edge inference engine. Convolutional layers extract spatial features across subcarriers, Gated Recurrent Units manage the temporal flow to maintain faster speeds on constrained hardware, and an attention mechanism dynamically weights critical impact moments within the signal.
- **LLM Reasoning Algorithms:** Process the transcribed text of the user's voice to evaluate linguistic coherence and emotional sentiment, making an autonomous classification regarding the validity of the emergency.

4.5 Project scheduling and tracking using Timeline / Gantt Chart

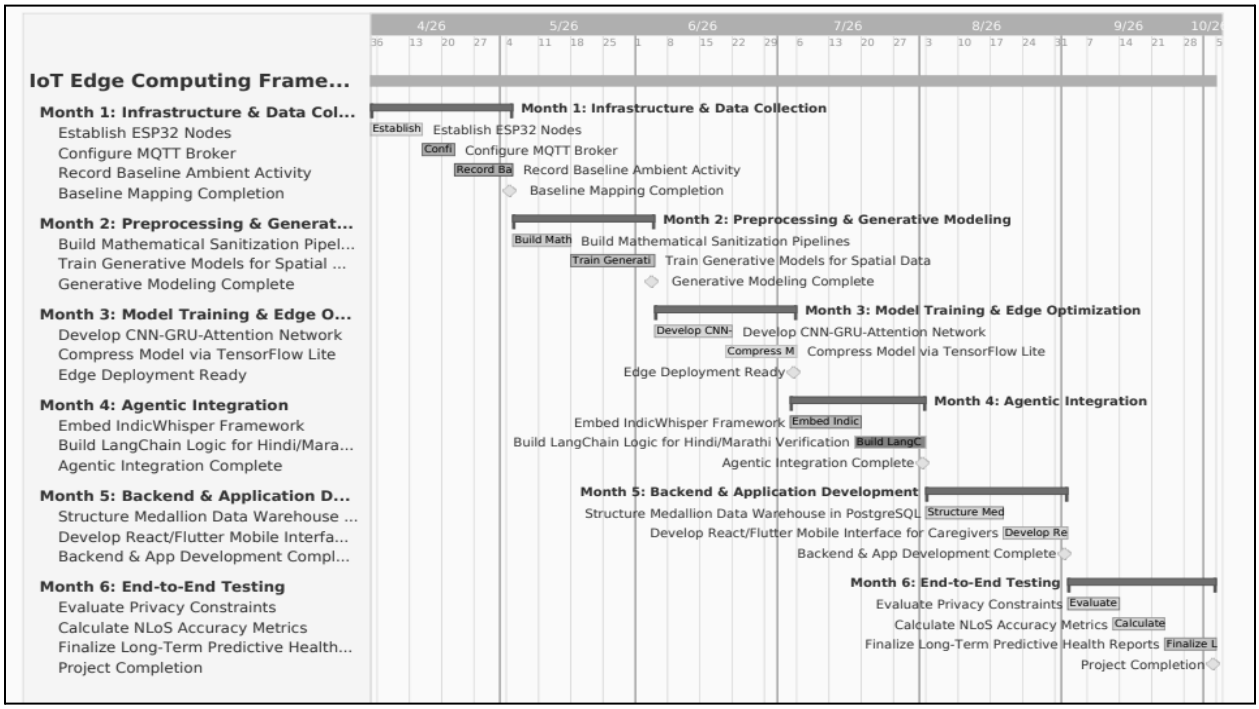


Fig. 4.5.1 Gantt Chart

The execution of the proposed framework is structured across a detailed six-month timeline:

- **Month 1 (Infrastructure & Data Collection):** Establish the ESP32 nodes, configure the MQTT broker, and record baseline ambient activity to map the standard signal environment.
- **Month 2 (Preprocessing & Generative Modeling):** Build the mathematical sanitization pipelines and train the generative models to synthesize diverse spatial datasets.
- **Month 3 (Model Training & Edge Optimization):** Develop the CNN-GRU-Attention network and systematically compress it via TensorFlow Lite for stable edge deployment.
- **Month 4 (Agentic Integration):** Embed the IndicWhisper framework and build the LangChain logic required for the two-way Hindi and Marathi verification system.
- **Month 5 (Backend & Application Development):** Structure the Medallion data warehouse in PostgreSQL and code the React/Flutter mobile interface for caregivers.
- **Month 6 (End-to-End Testing):** Evaluate privacy constraints, calculate NLoS accuracy metrics, and finalize the generation of long-term predictive health reports.

5. Implementation Results and Discussions

This chapter provides a detailed overview of the initial technical execution of the Sannidhi system. It covers the methods used for acquiring WiFi CSI data, the preprocessing techniques required to clean raw RF signals, the experimental setup of the sensing nodes, and the metrics used to evaluate the system's early-stage performance. This implementation phase bridges the gap between theoretical RF propagation and a functional proactive care model by establishing a robust, edge-based pipeline for real-time anomaly detection.

5.1 Data Collection / Acquisition

- **WiFi CSI Harvesting:** Utilizing ESP32-WROOM nodes, we extract raw Channel State Information (CSI) from the 2.4GHz WiFi subcarriers to act as an ambient RF sensor.
- **Continuous Capture:** The system performs continuous WiFi CSI capture, logging data across dimensions of time steps, subcarriers, and antennas to create a "physical tripwire".
- **Sample Scenarios:** Data is rigorously collected across four primary scenarios to build a baseline: Empty Room, Walking, Sitting, and Controlled Falls.
- **Environmental Simulation:** Testing is conducted in indoor residential spaces to simulate real-world multipath distortions and ensure the system can "see" through walls.

5.2 Data Preprocessing Techniques

- **Signal Conditioning:** Raw CSI data is highly susceptible to environmental noise; therefore, we implement Phase Sanitization and Noise Suppression on the local edge host.
- **Digital Filtering:** We utilize Kalman Filters and low-pass filtering to smooth signal amplitude and remove interference from other household electronic devices.
- **Feature Transformation:** Time-series arrays are converted into CSI Spectrograms or time-frequency representations to highlight the specific "ripples" caused by human kinetic energy.

5.3 Experimental Setup

- **Node Configuration:** A static pair of ESP32 modules (Transmitter and Receiver) is strategically placed to create an invisible sensing zone across a private room.
- **Edge Hub Integration:** The receiver node is tethered to a Raspberry Pi 5 (Local AI Hub) which runs the CNN-GRU-Attention Network entirely locally to ensure 100% privacy.
- **Hardware Validation:** Initial tests used a FreeRTOS task to track signal strength (RSSI) attenuation, proving that human movement causes a measurable drop in ambient WiFi signals.

5.4 Evaluation Metrics

- **Activity Recognition Accuracy:** The primary target is >90% accuracy for fall detection, particularly in Non-Line-of-Sight (NLoS) environments like bathrooms.
- **Agentic Proactivity Index:** This measures the system's ability to autonomously initiate a multilingual voice check-in (Hindi/Marathi) within a sub-5-second window of a detected anomaly.
- **Latency Benchmarking:** We monitor end-to-end processing time—from the physical fall to the caregiver’s SOS alert—to ensure local edge processing avoids slow cloud round-trips.

5.5 Gap Analysis of Existing Systems

- **Vs. Wearables:** Sannidhi eliminates the "compliance burden" and "device fatigue" where seniors frequently forget to wear, charge, or maintain their devices.
- **Vs. Cameras:** Our RF-based approach provides 100% privacy preservation in sensitive areas where cameras are rejected due to their intrusive "surveillance feel".
- **Vs. Passive AI:** Unlike dormant systems that require a "wake word," Sannidhi’s Agentic AI core uses physical signal triggers to autonomously verify emergencies.
- **Linguistic Inclusion:** While many systems are English-centric, Sannidhi provides critical support for Hindi and Marathi, removing the "technological exclusion" for the Indian geriatric demographic.

6. Plan of Action for the Next Semester

This chapter details the current development status of the Sannidhi project and outlines the specific milestones for Project Phase II. The plan focuses on moving from a proof-of-concept prototype to a fully integrated, scalable healthcare solution.

6.1 Work Done Till Date

- **Foundational Research:** Completed a comprehensive literature survey to identify critical gaps in "Passive AI" and camera-based systems, specifically focusing on the privacy-safety paradox.
- **Hardware Validation:** Successfully established a CSI capture link between ESP32 nodes and verified that physical movement causes measurable ripples in the WiFi signal.
- **Architectural Design:** Finalized the STAA Methodology (Sense-Think-Act-Analyze) and mapped it to a four-layer backend infrastructure.
- **UI/UX Prototyping:** Developed high-fidelity Figma mockups for the Caregiver Mobile App and the Clinical Health Dashboard to visualize patient status and long-term trends.

6.2 Plan of Action for Project II

- **Synthetic Data Generation :** Implement a cWGAN-GP (Conditional Wasserstein GAN) architecture to generate synthetic CSI samples. This will augment the dataset to include rare fall scenarios and diverse body types without requiring exhaustive physical testing.
- **DL Model Optimization :** Refine the CNN-GRU-AttNet (Attention Network) to improve fall detection precision and decrease latency during edge inference.
- **Multilingual NLP Integration:** Fully deploy Indic-Whisper locally on the Raspberry Pi 5 to support proactive, empathetic check-ins in Hindi and Marathi, removing the "technological exclusion" for the Indian demographic
- **Medallion DWH Deployment :** Build the full Bronze (Raw), Silver (Structured), and Gold (Analytical) data pipelines in PostgreSQL. This will enable the generation of weekly probabilistic reports on behavioral trends like anxiety and isolation.
- **Security & Privacy Hardening:** Integrate Differential Privacy and Federated Learning frameworks to ensure all sensitive patient interaction logs stay secured on the local host.

- **System Integration & Pilot Study:** Interface the hardware nodes with the React/Flutter mobile app for a 2-week field test. This pilot will be used to refine the "Proactivity Index" and measure end-to-end system response times.
- **Strategic Roadmap & Enterprise Scaling:**
    - B2C Peace of Mind: The immediate goal is to provide affordable, invisible safety for Indian middle-class families through a subscription-based caregiver alert model.
    - B2B Healthcare Integration: Our long-term roadmap includes integrating the Sannidhi AI engine with Smart-Home ecosystems and Home Healthcare agencies to create a ubiquitous, nationwide safety net.
    - Empowering Aging-in-Place: By transforming independent living from a state of vulnerability into a secure, dignified experience, Sannidhi establishes a scalable blueprint for Geriatrics Engineering.

Phase	**Timeline**	**Task Description & Deliverables**
1. Research	Weeks 1-4	Literature survey on CSI Sensing, cWGAN-GP, and Medallion DWH. Del: Survey Paper.
2. Data Engine	Weeks 5-10	Baseline collection via ESP32 nodes; synthetic data generation via cWGAN-GP. Del: Augmented CSI Dataset.
3.Model Development	Weeks 11-16	Training CNN-GRU-AttNet (Edge) and Indic-Whisper LLM (NLP Agent). Del: Trained AI Models.
4. Data Infrastructure	Weeks 17-20	Implementing Medallion DWH for secure refinement and health tracking. Del: ETL Pipeline.
5. Integration	Weeks 21-24	Interfacing Raspberry Pi, ESP32 sensors, and React/Flutter Mobile App. Del: Functional Prototype.
6. Validation	Weeks 25-28	Performance benchmarking (Latency/Accuracy) and final documentation. Del: Final Report.

Table 6.2.1 Plan of Action And Deliverables

7. Conclusion

The exponential rise in the global geriatric population, coupled with a shrinking caregiver workforce, has exposed severe vulnerabilities in current eldercare monitoring. Traditional systems—relying either on privacy-violating cameras or high-friction wearable devices—fail to provide an acceptable balance of patient dignity and proactive safety. Sannidhi was conceptualized to bridge this critical healthcare gap by transforming ambient radio frequencies into an invisible, context-aware safety net.

Throughout the first phase of this project, a robust, privacy-preserving edge-computing architecture was designed and implemented. By engineering a distributed hardware network, the system integrates advanced WiFi Channel State Information (CSI) sensing, optimized Deep Learning sequence models (CNN-GRU-AttNet), and a localized Agentic AI core. The implementation of the Sense-Think-Act-Analyze (STAA) methodology and localized, multilingual voice verification (Indic-Whisper) bypassed the high false-positive rates typical of passive radars, resulting in a highly efficient, closed-loop triage pipeline that executes entirely on the edge to guarantee data privacy.

Moving into the next phase of development, the project will transition from a foundational processing engine into a fully integrated, scalable health informatics solution. The planned integration of generative data augmentation (cWGAN-GP), a structured Medallion Data Warehouse, and a cross-platform caregiver interface will allow the system to map long-term behavioral trends and generate predictive mobility reports. Furthermore, the introduction of enterprise scaling roadmaps will ensure the platform's viability within formal home healthcare ecosystems by providing vital analytical insights to medical professionals and caregivers.

Ultimately, Sannidhi represents a paradigm shift in geriatrics engineering by moving beyond reactive alarms to deliver proactive, empathetic care. Leveraging device-free ambient intelligence empowers the elderly with true independent living, directly advancing the UN Sustainable Development Goals of Good Health and Well-being (SDG 3), Industry, Innovation and Infrastructure (SDG 9), Reduced Inequalities (SDG 10), Sustainable Cities and Communities (SDG 11), and Peace, Justice and Strong Institutions (SDG 16).

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9. Appendix

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